HONORS CHEMISTRY • UNIT 6

# Stoichiometry

Use a balanced equation like a recipe: predict how much you need, how much you make, what runs out first, and how well the lab went.

TEACHER EDITION

Everything in a light-gray box or cell is an answer or model response. Teacher notes (misconceptions, demo ideas, discussion prompts, sources) appear in the heavy-bordered boxes labeled TEACHER NOTE. The pages follow the student packet, so they line up closely.

## What we will learn

| | |
|---|---|
| **6.1** | Mole ratios |
| **6.2** | Mass-to-mass stoichiometry |
| **6.3** | Liters and particles (gas volume at STP) |
| **6.4** | Limiting and excess reactants (BCA tables) |
| **6.5** | Theoretical, actual and percent yield |

## By the end of this unit I can...

- Write mole ratios from a balanced equation and use them to convert between moles of two substances.
- Solve mass-to-mass problems using a mole map and show every unit.
- Convert between liters of gas at STP, moles, grams and particles.
- Identify the limiting reactant and calculate product made and excess left over using a BCA table.
- Calculate theoretical yield and percent yield, and explain why actual yield is smaller.

TEACHER OVERVIEW

**Numbering:** the homework folder labels these Topics 6.1–6.3, 5.4 and 5.5 (the last two are mis-numbered in the PDF). This packet is Unit 6 in the notes sequence with topics 6.1–6.5.

**Every answer is computed** by code from the balanced equation, using 2-decimal atomic masses (H 1.01, C 12.01, N 14.01, O 16.00, Na 22.99, Al 26.98, P 30.97, S 32.06, Cl 35.45, K 39.10, Fe 55.85, Ag 107.87, Ca 40.08) and 22.4 L/mol at STP. Unrounded values carry through; only the final answer is rounded to the sig figs of the given data. If your class uses different masses the last digit may differ.

**Homework quirks:** 6.1 #5 writes hydrazine as $N_2H_2$ (real hydrazine is $N_2H_4$; the packet keeps the homework formula so the key matches). 6.3 #3 calls $NaN_3$ "sodium nitride" (it is sodium azide). The 6.3 sub-parts are lettered b, e–j in the PDF; the packet renumbers them. 6.3 #1 needs STP assumed. 6.5 #2 gives "10 grams" of $AgNO_3$, treated as 2 s.f. 6.5 #3 as written (11.3 g actual) gives a percent yield over 100 % (theoretical is 8.78 g), so the packet uses 7.30 g.

**Misconceptions:** using the coefficients as grams; skipping the mole step; picking the limiting reactant as the one with the smaller mass; dividing by the coefficient in the wrong direction; reporting excess in grams when the question asks for moles or liters; calling percent yield > 100 % acceptable.

**Pacing idea:** 6.1–6.2 on day 1 (mole map until students can draw it from memory), 6.3 on day 2 (STP lab video), 6.4 on days 3–4 (s'mores first, then BCA), 6.5 plus practice on day 5.

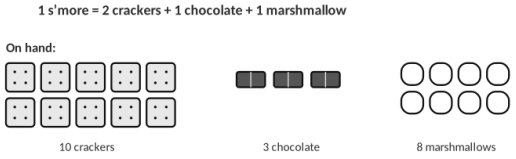
# 6.1 Mole Ratios *The coefficients of a balanced equation are a recipe*

BIG IDEA

A balanced equation tells you how many **particles** react, and since a mole is a counting number, the coefficients also give **moles**. The **mole ratio** is a conversion factor built from two coefficients. It is the only bridge between one substance and another.

**Key vocabulary — write each definition in your own words**

| Term | Definition (your words) | Example / sketch |
|---|---|---|
| **Stoichiometry** | The calculation of amounts of reactants and products in a chemical reaction using a balanced equation. | How much $NH_3$ forms from 6 mol $H_2$? |
| **Mole ratio** | A ratio between the moles of any two substances in a balanced equation, taken from their coefficients. | $N_2 : H_2 = 1 : 3$ |




*Analogy: the recipe 2 crackers : 1 chocolate : 1 marshmallow plays the role of the coefficients.*

**1.** Use the s'more recipe. How many s'mores can you make from 6 chocolates if you have plenty of everything else? How many crackers does that use? Write the two ratios you used.

6 s'mores (1 chocolate : 1 s'more) using 12 crackers (2 crackers : 1 chocolate). The ratios are 1 s'more / 1 chocolate and 2 crackers / 1 chocolate.

**2.** What do you need before you can write a mole ratio? Why must the equation be balanced first?

A balanced chemical equation. The coefficients only equal the true particle ratio when the equation is balanced; an unbalanced equation would give the wrong ratio.

**Write the mole ratios**

| Equation | Ratios (write each as a fraction) |
|---|---|
| $N_2 + 3H_2 \rightarrow 2NH_3$ | $N_2/H_2$ = _____ $N_2/NH_3$ = _____ $H_2/NH_3$ = _____ |
| $8H_2 + S_8 \rightarrow 8H_2S$ | $H_2/H_2S$ = _____ $H_2/S_8$ = _____ $H_2S/S_8$ = _____ |

ANSWERS

$N_2/H_2 = 1/3$; $N_2/NH_3 = 1/2$; $H_2/NH_3 = 3/2$. $H_2/H_2S = 8/8 = 1/1$; $H_2/S_8 = 8/1$; $H_2S/S_8 = 8/1$.

**3.** Hydrogen gas reacts with oxygen gas to make water. (a) Write the balanced equation. (b) Write the ratios $H_2/O_2$, $H_2/H_2O$ and $O_2/H_2O$. (c) How many moles of water form from 10 moles of hydrogen?

**a.** Balanced equation

$2H_2\ (g) + O_2\ (g) \rightarrow 2H_2O\ (l)$

**Answer (units & sig figs):** **$2H_2 + O_2 \rightarrow 2H_2O$**

**b.** Mole ratios

$H_2/O_2 = 2/1$; $H_2/H_2O = 2/2 = 1/1$; $O_2/H_2O = 1/2$

**Answer (units & sig figs):**

**c.** Moles of water

$10\ mol\ H_2 \times (2\ mol\ H_2O / 2\ mol\ H_2) = 1 \times 10^1\ mol\ H_2O$

**Answer (units & sig figs):** **$1 \times 10^1$ mol $H_2O$**

**WORKED EXAMPLE — MOLE RATIO**

How many moles of $NH_3$ form from 4.0 mol $N_2$? $N_2 + 3H_2 \rightarrow 2NH_3$

$4.0\ mol\ N_2 \times (2\ mol\ NH_3 / 1\ mol\ N_2) = 8.0\ mol\ NH_3$

**Answer:** **8.0 mol $NH_3$**

**4.** Use $N_2 + 3H_2 \rightarrow 2NH_3$.

**a.** If 10 moles of $NH_3$ are produced, how many moles of $N_2$ were needed?

$10\ mol\ NH_3 \times (1\ mol\ N_2 / 2\ mol\ NH_3) = 5\ mol\ N_2$

**Answer (units & sig figs):** **5 mol $N_2$**

**b.** If 3.00 moles of $H_2$ are used, how many moles of $NH_3$ are made?

$3.00\ mol\ H_2 \times (2\ mol\ NH_3 / 3\ mol\ H_2) = 2.00\ mol\ NH_3$

**Answer (units & sig figs):** **2.00 mol $NH_3$**

**c.** If 0.600 moles of $NH_3$ are produced, how many moles of $H_2$ are required?

$0.600\ mol\ NH_3 \times (3\ mol\ H_2 / 2\ mol\ NH_3) = 0.900\ mol\ H_2$

**Answer (units & sig figs):** **0.900 mol $H_2$**

**5.** The reaction between hydrazine ($N_2H_2$) and hydrogen peroxide ($H_2O_2$) fuels rockets. The products are nitrogen gas and water. (a) Write the balanced equation. (b) How many moles of $N_2$ form from 37 moles of hydrazine? (c) If 23 moles of water form, how many moles of $H_2O_2$ were used?

**a.** Balanced equation

$N_2H_2$ (l) + $H_2O_2$ (l) → $N_2$ (g) + 2$H_2O$ (g)

**Answer (units & sig figs):**

**b.** Moles of $N_2$

37 mol $N_2H_2$ × (1 mol $N_2$ / 1 mol $N_2H_2$) = 37 mol $N_2$

**Answer (units & sig figs):** **37 mol $N_2$**

**c.** Moles of $H_2O_2$

23 mol $H_2O$ × (1 mol $H_2O_2$ / 2 mol $H_2O$) = 12 mol $H_2O_2$

**Answer (units & sig figs):** **12 mol $H_2O_2$**

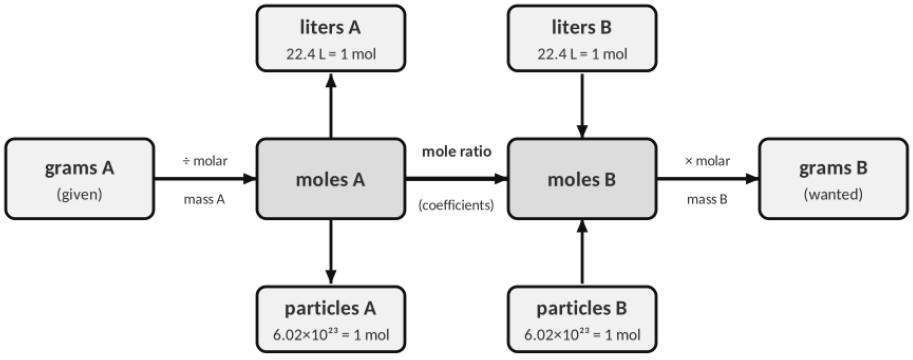
**TEACHER NOTE**

The homework writes hydrazine as $N_2H_2$; real hydrazine is $N_2H_4$ ($N_2H_4$ + 2$H_2O_2$ → $N_2$ + 4$H_2O$, so 23 mol $H_2O$ would need 11.5 mol $H_2O_2$). The key above uses the homework formula.

## 6.2 Mass-to-Mass Stoichiometry *Grams → moles → moles → grams*

**BIG IDEA**

You cannot compare grams of different substances directly, because a gram of one is not the same number of particles as a gram of another. Convert to **moles**, use the **mole ratio**, then convert back to **grams**.




*The mole map (use the top and bottom boxes in 6.3). Molar masses come from the periodic table.*

**1.** In your own words, why is the mole ratio step the only step that uses the balanced equation? What do the other two steps use?

The mole ratio compares particles of two different substances, and only the coefficients tell us that. Converting grams to moles and back uses only each substance's own molar mass.

**SETUP RULES**

1. Balance the equation. 2. Write the given quantity with its unit and substance. 3. Chain factors so that each unit cancels. 4. Round the final answer to the sig figs of the given data.

**WORKED EXAMPLE — GRAMS TO GRAMS**

How many grams of FeS form from 5.9 g of Fe? ($8Fe + S_8 \rightarrow 8FeS$)

5.9 g Fe × (1 mol Fe / 55.85 g) × (8 mol FeS / 8 mol Fe) × (87.91 g FeS / 1 mol) = 9.3 g FeS

**Answer:** **9.3 g FeS**

**2.** Use the equation $Fe + S_8 \rightarrow FeS$. (a) Balance it.

**a.** Balance the equation

$8Fe + S_8 \rightarrow 8FeS$

**Answer (units & sig figs):** **$8Fe + S_8 \rightarrow 8FeS$**

**b.** How many moles of FeS are produced when 2.4 g of $S_8$ react?

2.4 g $S_8$ × (1 mol $S_8$ / 256.48 g) × (8 mol FeS / 1 mol $S_8$) = 0.075 mol FeS

**Answer (units & sig figs):** **0.075 mol FeS**

**c.** How many grams of iron are used if 11.9 g of sulfur ($S_8$) are used?

11.9 g $S_8$ × (1 mol $S_8$ / 256.48 g) × (8 mol Fe / 1 mol $S_8$) × (55.85 g Fe / 1 mol) = 20.7 g Fe

**Answer (units & sig figs):** **20.7 g Fe**

**3.** Use the equation $Fe_2O_3 + C \rightarrow Fe + CO_2$ (the reaction in a blast furnace).

**a.** Balance the equation

$2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$

**Answer (units & sig figs):** **$2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$**

**b.** How many grams of iron form if 39.64 g of carbon are used?

39.64 g C × (1 mol C / 12.01 g) × (4 mol Fe / 3 mol C) × (55.85 g Fe / 1 mol) = 245.8 g Fe

**Answer (units & sig figs):** **245.8 g Fe**

**c.** If 63.12 g of iron(III) oxide are used, how many grams of carbon dioxide form?

63.12 g $Fe_2O_3$ × (1 mol $Fe_2O_3$ / 159.70 g) × (3 mol $CO_2$ / 2 mol $Fe_2O_3$) × (44.01 g $CO_2$ / 1 mol) = 26.09 g $CO_2$

**Answer (units & sig figs):** **26.09 g $CO_2$**

**4.** A student solved "grams of $Fe_2O_3$ → grams of $CO_2$" by multiplying by 159.7 g/mol and then by 44.01 g/mol, skipping the mole ratio. Describe what is wrong and what the student's answer would represent.

The student never converted grams of $Fe_2O_3$ to moles and never used the mole ratio (2 $Fe_2O_3$ : 3 $CO_2$), so the units do not cancel. Grams of one substance cannot be converted directly into grams of another; the coefficients only connect moles.

**5.** To convert grams of A to grams of B you need…

A) molar masses only **B) molar masses and the mole ratio** C) the mole ratio only D) 22.4 L/mol

## 6.3 Liters and Particles *Gas volume at STP and counting atoms*

**BIG IDEA**

At **STP** (0 °C and 1 atm), **1 mole of any gas occupies 22.4 L**. This lets us convert between liters of a gas and moles, and so to any other substance. One mole is also $6.02 \times 10^{23}$ particles.

**Key vocabulary — write each definition in your own words**

| Term | Definition (your words) | Example / sketch |
|---|---|---|
| **STP** | Standard temperature and pressure: 0 °C and 1 atm; 1 mol of any gas occupies 22.4 L. | 5.60 L of $O_2$ = 0.250 mol |
| **Molar volume** | The volume of one mole of gas (22.4 L/mol at STP); it is the same for every ideal gas. | 22.4 L/mol |

**THINK OF IT LIKE… A DOZEN IS JUST A NUMBER**

A dozen is 12 things, whether they are eggs or elephants. A mole is $6.02 \times 10^{23}$ things, and at STP that many gas molecules take up 22.4 L whether they are $H_2$ or $Cl_2$. Gases are mostly empty space, so the size of the molecule hardly matters.

**1.** Explain why 1 mole of $H_2$ and 1 mole of $Cl_2$ occupy the same volume at STP even though a $Cl_2$ molecule is much heavier.

At STP the gas molecules are far apart (mostly empty space), so the volume depends on the number of particles, not on their size or mass. Equal numbers of particles occupy equal volumes.

**2.** Hydrogen gas and oxygen gas combine to form water. Assume STP.

**a.** How many liters of $H_2$ are needed to make 3 moles of water?

$3 \text{ mol } H_2O \times (2 \text{ mol } H_2 / 2 \text{ mol } H_2O) \times (22.4 \text{ L } H_2 / 1 \text{ mol}) = 7 \times 10^1 \text{ L } H_2$

**Answer (units & sig figs):** **$7 \times 10^1$ L $H_2$**

**b.** How many grams of water form from 30.2 L of hydrogen?

$30.2 \text{ L } H_2 \times (1 \text{ mol } H_2 / 22.4 \text{ L}) \times (2 \text{ mol } H_2O / 2 \text{ mol } H_2) \times (18.02 \text{ g } H_2O / 1 \text{ mol}) = 24.3 \text{ g } H_2O$

**Answer (units & sig figs):** **24.3 g $H_2O$**

**c.** How many grams of water form from 30.2 L of oxygen?

$30.2 \text{ L } O_2 \times (1 \text{ mol } O_2 / 22.4 \text{ L}) \times (2 \text{ mol } H_2O / 1 \text{ mol } O_2) \times (18.02 \text{ g } H_2O / 1 \text{ mol}) = 48.6 \text{ g } H_2O$

**Answer (units & sig figs):** **48.6 g $H_2O$**

**d.** When 120.3 g of water form, how many liters of hydrogen were used?

$120.3 \text{ g } H_2O \times (1 \text{ mol } H_2O / 18.02 \text{ g}) \times (2 \text{ mol } H_2 / 2 \text{ mol } H_2O) \times (22.4 \text{ L } H_2 / 1 \text{ mol}) = 149.5 \text{ L } H_2$

**Answer (units & sig figs):** **149.5 L $H_2$**

## LAB CORNER • Sodium azide in a car air bag

A crash sensor sets off the reaction $NaN_3(s) \rightarrow Na(l) + N_2(g)$. The nitrogen gas inflates the bag in about 0.03 seconds.

[ IMAGE GOES HERE ]

*Photo or diagram of an air bag inflator (add your own)*

**3.** Balance the equation, then solve (STP).

**a.** Balance the equation

$2NaN_3\ (s) \rightarrow 2Na\ (l) + 3N_2\ (g)$

**Answer (units & sig figs):** **$2NaN_3\ (s) \rightarrow 2Na\ (l) + 3N_2\ (g)$**

**b.** How many liters of $N_2$ form from 15.62 g of $NaN_3$?

$15.62\ g\ NaN_3 \times (1\ mol\ NaN_3 / 65.02\ g) \times (3\ mol\ N_2 / 2\ mol\ NaN_3) \times (22.4\ L\ N_2 / 1\ mol) = 8.072\ L\ N_2$

**Answer (units & sig figs):** **8.072 L $N_2$**

**c.** How many grams of $NaN_3$ are needed to produce 200.0 L of $N_2$?

$200.0\ L\ N_2 \times (1\ mol\ N_2 / 22.4\ L) \times (2\ mol\ NaN_3 / 3\ mol\ N_2) \times (65.02\ g\ NaN_3 / 1\ mol) = 387.0\ g\ NaN_3$

**Answer (units & sig figs):** **387.0 g $NaN_3$**

**d.** If 31.4 g of sodium are produced, how many liters of $N_2$ are produced?

$31.4\ g\ Na \times (1\ mol\ Na / 22.99\ g) \times (3\ mol\ N_2 / 2\ mol\ Na) \times (22.4\ L\ N_2 / 1\ mol) = 45.9\ L\ N_2$

**Answer (units & sig figs):** **45.9 L $N_2$**

**TEACHER NOTE**

The homework says "sodium nitride"; $NaN_3$ is sodium azide.

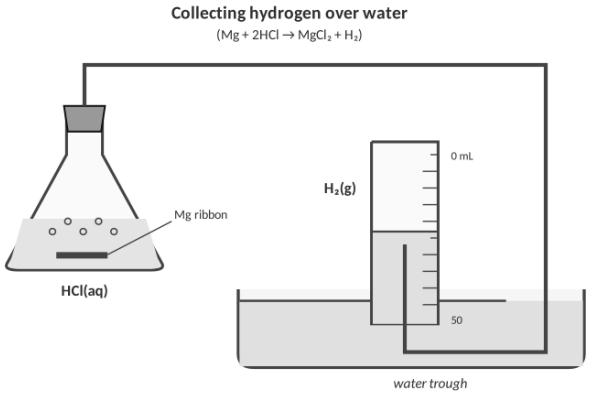
**4.** How many molecules of water form from 5.0 g of hydrogen gas? ($2H_2 + O_2 \rightarrow 2H_2O$; 1 mol = $6.02 \times 10^{23}$ molecules)

$5.0\ g\ H_2 \times (1\ mol\ H_2 / 2.02\ g) \times (2\ mol\ H_2O / 2\ mol\ H_2) \times (6.02 \times 10^{23}\ molecules / 1\ mol) = 1.5 \times 10^{24}\ molecules\ H_2O$

**Answer (units & sig figs):** **$1.5 \times 10^{24}$ molecules $H_2O$**

## LAB CORNER • Hydrogen gas collected over water

A 0.150 g strip of magnesium reacts completely with excess hydrochloric acid. The hydrogen is collected in an inverted graduated cylinder.




**5.** (a) Write the balanced equation. (b) Calculate the theoretical volume of $H_2$ at STP in mL. (c) The measured volume was 128 mL. Name one reason it differs.

**a.** Equation

Mg (s) + 2HCl (aq) → $MgCl_2$ (aq) + $H_2$ (g)

**Answer (units & sig figs):** **Mg (s) + 2HCl (aq) → $MgCl_2$ (aq) + $H_2$ (g)**

**b.** Theoretical volume

0.150 g Mg × (1 mol Mg / 24.31 g) × (1 mol $H_2$ / 1 mol Mg) × (22.4 L $H_2$ / 1 mol) = 0.138 L = 138 mL

**Answer (units & sig figs):** **138 mL $H_2$**

**c.** Reason

Some gas stays dissolved or in the tubing; the room is not at STP (T and P differ); water vapor is mixed in with the $H_2$; the Mg may not be pure.

**Answer (units & sig figs):**

# 6.4 Limiting and Excess Reactants *Which ingredient runs out first?*

**BIG IDEA**

Reactants are almost never mixed in perfect recipe amounts. The **limiting reactant** is the one that runs out first and so decides how much product forms. The **excess reactant** is what is left over when the reaction stops.

**Key vocabulary — write each definition in your own words**

| Term | Definition (your words) | Example / sketch |
|---|---|---|
| **Limiting reactant** | The reactant that is completely used up first; it determines the maximum amount of product. | Chocolate in the s'more example |
| **Excess reactant** | The reactant present in more than the amount needed; some is left over after the reaction. | Crackers and marshmallows |




*Recipe: 2 crackers + 1 chocolate + 1 marshmallow → 1 s'more.*

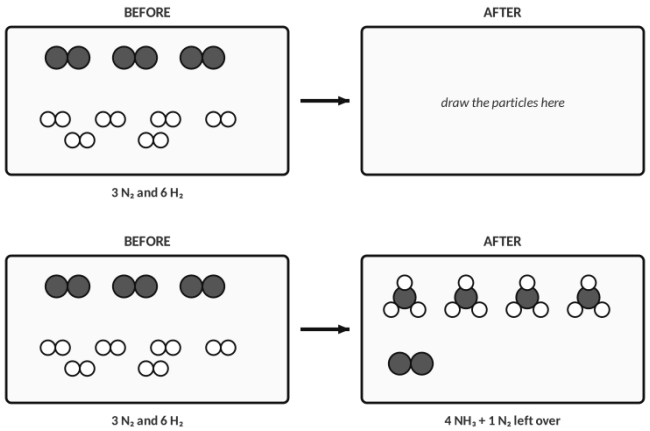
**1.** Using the s'more recipe and the items shown: (a) How many s'mores can you make? (b) What is the limiting "reactant"? (c) What is left over, and how many of each?

(a) Crackers allow 10 ÷ 2 = 5; chocolate allows 3; marshmallows allow 8. The smallest is 3, so 3 s'mores.
(b) Chocolate is limiting.
(c) Crackers used 6, so 4 left; marshmallows used 3, so 5 left. Chocolate: 0 left.

**BCA TABLE STEPS (MOLES!)**

1. Balance the equation. Convert every given amount to **moles**.
2. Divide each by its coefficient; the **smallest** value is the limiting reactant.
3. **Before** row = starting moles. **Change** row = the limiting reactant goes to 0; use the mole ratio for the rest (reactants −, products +).
4. **After** row = Before + Change. Convert the After amounts to the unit the question asks for.

**Particle picture: $N_2 + 3H_2 \rightarrow 2NH_3$**




*Answer: 6 $H_2$ react with 2 $N_2$ to make 4 $NH_3$; hydrogen is limiting and 1 $N_2$ is left over.*

**2.** Which reactant is limiting? Draw the particles after the reaction in the AFTER box, then say how many $N_2$ molecules are left.

$H_2$ is limiting (6 $H_2$ need only 2 $N_2$; 3 $N_2$ are available). After: 4 $NH_3$ and 1 $N_2$ left over.

**WORKED EXAMPLE — BCA TABLE**

1.5 mol $CH_4$ react with 2.5 mol $Cl_2$: $CH_4 + 2Cl_2 \rightarrow CH_2Cl_2 + 2HCl$. Find the moles of each product and the moles of excess reactant left.

| (mol) | $CH_4$ | $2Cl_2$ | $CH_2Cl_2$ | 2HCl |
|---|---|---|---|---|
| **Before** | 1.50 | 2.50 | 0 | 0 |
| **Change** | -1.25 | -2.50 | +1.25 | +2.50 |
| **After** | 0.250 | 0 | 1.25 | 2.50 |

Limiting reactant: $Cl_2$ (it gives the smallest amount of reaction, 1.3 "equation units").
$CH_2Cl_2$ produced: 1.3 mol = 1.3 mol
HCl produced: 2.5 mol = 2.5 mol
$CH_4$ left over: 0.25 mol = 0.25 mol

**3.** 15.0 g of $N_2$ gas reacts with 4.50 g of $H_2$ gas: $N_2 + 3H_2 \rightarrow 2NH_3$. Use a BCA table to find the mass of $NH_3$ made and the mass of the excess reactant left.

| (mol) | $N_2$ | $3H_2$ | $2NH_3$ |
|---|---|---|---|
| **Before** | 0.535 | 2.23 | 0 |
| **Change** | -0.535 | -1.61 | +1.07 |
| **After** | 0 | 0.622 | 1.07 |

Limiting reactant: $N_2$ (it gives the smallest amount of reaction, 0.535 "equation units").
$NH_3$ produced: 1.07 mol = 18.2 g
$N_2$ left over: 0 mol = 0 g
$H_2$ left over: 0.622 mol = 1.26 g

**4.** 17.5 L of $NH_3$ gas react with 12.5 L of $O_2$ gas at STP: $4NH_3\ (g) + 5O_2\ (g) \rightarrow 4NO\ (g) + 6H_2O\ (g)$. Use a BCA table to find the mass of each product and the liters of excess reactant left.

| (mol) | $4NH_3$ | $5O_2$ | 4NO | $6H_2O$ |
|---|---|---|---|---|
| **Before** | 0.781 | 0.558 | 0 | 0 |
| **Change** | -0.446 | -0.558 | +0.446 | +0.670 |
| **After** | 0.335 | 0 | 0.446 | 0.670 |

Limiting reactant: $O_2$ (it gives the smallest amount of reaction, 0.112 "equation units").
NO produced: 0.446 mol = 13.4 g
$H_2O$ produced: 0.670 mol = 12.1 g
$NH_3$ left over: 0.335 mol = 7.50 L

**5.** 7.50 g of solid sodium react with 18.5 L of $Cl_2$ gas at STP: $2Na\ (s) + Cl_2\ (g) \rightarrow 2NaCl\ (s)$. Find the mass of NaCl made and the mass of the excess reactant left.

| (mol) | 2Na | $Cl_2$ | 2NaCl |
|---|---|---|---|
| **Before** | 0.326 | 0.826 | 0 |
| **Change** | -0.326 | -0.163 | +0.326 |
| **After** | 0 | 0.663 | 0.326 |

Limiting reactant: Na (it gives the smallest amount of reaction, 0.163 "equation units").
NaCl produced: 0.326 mol = 19.1 g
$Cl_2$ left over: 0.663 mol = 47.0 g

**6.** A classmate says, "The limiting reactant is the one with the smaller mass." Use any example above to show why this is not reliable.

Limiting depends on moles divided by coefficient, not mass. In the $N_2/H_2$ problem $H_2$ has the smaller mass (4.50 g vs 15.0 g) and is limiting, but in the sodium/chlorine problem the $Cl_2$ has far more mass (47 g vs 7.5 g) and is the excess. Different substances have different molar masses and different coefficients.

**7.** In a BCA table, the limiting reactant's "After" value is always…

A) equal to its Before value **B) zero** C) equal to the product D) negative

## 6.5 Theoretical, Actual and Percent Yield *How well did the lab go?*

**BIG IDEA**

The **theoretical yield** is the most product the equation predicts (from the limiting reactant). The **actual yield** is what you really collect in the lab. **Percent yield = actual ÷ theoretical × 100**, and it can never honestly be more than 100 %.

**Key vocabulary — write each definition in your own words**

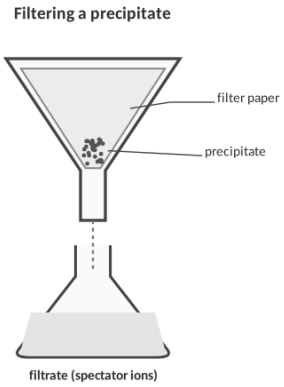
| Term | Definition (your words) | Example / sketch |
|---|---|---|
| **Theoretical yield** | The maximum amount of product calculated from the balanced equation and the limiting reactant. | 8.78 g $Al(OH)_3$ |
| **Actual yield** | The amount of product actually obtained and measured in an experiment. | 7.30 g $Al(OH)_3$ |
| **Percent yield** | The actual yield divided by the theoretical yield, times 100. | 7.30 / 8.78 × 100 = 83.1 % |

**THINK OF IT LIKE… FREE THROWS**

A player who shoots 10 free throws could, in theory, make all 10 (theoretical yield). Making 8 is the actual yield, so the percent yield is 80 %. Nobody can make 12 out of 10, so a percent yield over 100 % means something is wrong.

## LAB CORNER • Collecting a precipitate

A student mixes two solutions, filters the solid, dries it and finds its mass.




**1.** Give three reasons the actual yield of a precipitate is usually less than the theoretical yield. Then give one reason a student might get a percent yield greater than 100 %.

Less: some precipitate stays on the glassware or filter paper; some dissolves a little (not perfectly insoluble); incomplete reaction; side reactions; spills; impure reactants. More than 100 %: the solid was not completely dry (water mass counted) or was contaminated with another substance.

**2.** Methane burns: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$.

**a.** A student reacts 10.0 g of $O_2$ with excess $CH_4$. Calculate the theoretical yield of $CO_2$.

10.0 g $O_2$ × (1 mol $O_2$ / 32.00 g) × (1 mol $CO_2$ / 2 mol $O_2$) × (44.01 g $CO_2$ / 1 mol) = 6.88 g $CO_2$

**Answer (units & sig figs):** **6.88 g $CO_2$**

**b.** The lab produces 6.50 g of $CO_2$. Calculate the percent yield.

6.50 g / 6.88 g × 100 = 94.5 %

**Answer (units & sig figs):** **94.5 %**

**3.** $CaCl_2$ (aq) + $2AgNO_3$ (aq) → $Ca(NO_3)_2$ (aq) + 2AgCl (s)

**a.** A student reacts 15.0 g of $CaCl_2$ with 10. g of $AgNO_3$. Find the limiting reactant and the theoretical yield of AgCl.

Moles: $CaCl_2$ 0.135, $AgNO_3$ 0.059. $AgNO_3$ gives less (÷2), so it is limiting.
0.059 mol $AgNO_3$ × (2 mol AgCl / 2 mol $AgNO_3$) × (143.32 g / 1 mol) = 8.4 g

**Answer (units & sig figs):** **8.4 g AgCl**

**b.** 4.50 g of AgCl are obtained. Find the percent yield.

4.50 g / 8.4 g × 100 = 54 %

**Answer (units & sig figs):** **54 %**

**4.** $AlCl_3 + 3NaOH \rightarrow Al(OH)_3 + 3NaCl$

**a.** 15.0 g of $AlCl_3$ react with 20.0 g of NaOH. Find the limiting reactant and the theoretical yield of $Al(OH)_3$.

Moles: $AlCl_3$ 0.113, NaOH 0.500; ÷ coefficients: 0.113 vs 0.167. $AlCl_3$ is limiting.
0.113 mol × (1/1) × (78.01 g/mol) = 8.78 g

**Answer (units & sig figs):** **8.78 g $Al(OH)_3$**

**b.** 7.30 g of $Al(OH)_3$ are collected. Find the percent yield.

7.30 g / 8.78 g × 100 = 83.1 %

**Answer (units & sig figs):** **83.1 %**

**5.** A reaction has a theoretical yield of 25.0 g. A student collects 21.0 g. The percent yield is…

**A) 84.0 %** B) 119 % C) 16.0 % D) 46.0 g

PRACTICE

# Extra Practice — Unit 6

Show every setup with units.

## A. Mole ratios and mass-to-mass

**1.** Use $K_3PO_4 + Al(NO_2)_3 \rightarrow 3KNO_2 + AlPO_4$.

**a.** How many grams of $KNO_2$ form from 4.6 mol of $K_3PO_4$?

4.6 mol $K_3PO_4$ × (3 mol $KNO_2$ / 1 mol $K_3PO_4$) × (85.11 g $KNO_2$ / 1 mol) = $1.2 \times 10^3$ g $KNO_2$

**Answer (units & sig figs):** **$1.2 \times 10^3$ g $KNO_2$**

**b.** If 5.8 g of $AlPO_4$ form, how many grams of $Al(NO_2)_3$ were used?

5.8 g $AlPO_4$ × (1 mol $AlPO_4$ / 121.95 g) × (1 mol $Al(NO_2)_3$ / 1 mol $AlPO_4$) × (165.01 g $Al(NO_2)_3$ / 1 mol) = 7.8 g $Al(NO_2)_3$

**Answer (units & sig figs):** **7.8 g $Al(NO_2)_3$**

**c.** How many grams of $KNO_2$ form from 6.2 g of $K_3PO_4$?

6.2 g $K_3PO_4$ × (1 mol $K_3PO_4$ / 212.27 g) × (3 mol $KNO_2$ / 1 mol $K_3PO_4$) × (85.11 g $KNO_2$ / 1 mol) = 7.5 g $KNO_2$

**Answer (units & sig figs):** **7.5 g $KNO_2$**

**2.** Use $CO + 2H_2 \rightarrow CH_3OH$.

**a.** If 5.7 g of CO are used, how many grams of $H_2$ are also used?

5.7 g CO × (1 mol CO / 28.01 g) × (2 mol $H_2$ / 1 mol CO) × (2.02 g $H_2$ / 1 mol) = 0.82 g $H_2$

**Answer (units & sig figs):** **0.82 g $H_2$**

**b.** If 0.89 g of $H_2$ are used, how many grams of $CH_3OH$ form?

0.89 g $H_2$ × (1 mol $H_2$ / 2.02 g) × (1 mol $CH_3OH$ / 2 mol $H_2$) × (32.05 g $CH_3OH$ / 1 mol) = 7.1 g $CH_3OH$

**Answer (units & sig figs):** **7.1 g $CH_3OH$**

**c.** If 15.0 L of CO are used at STP, how many liters of $H_2$ are also used? Why is no molar mass needed?

15.0 L CO × (1 mol CO / 22.4 L) × (2 mol $H_2$ / 1 mol CO) × (22.4 L $H_2$ / 1 mol) = 30.0 L $H_2$

**Answer (units & sig figs):** **30.0 L $H_2$**

TEACHER NOTE

(c) Equal volumes of gases at the same T and P are equal numbers of moles, so liters convert with the coefficients alone (Avogadro). 15.0 L CO × (2 mol $H_2$/1 mol CO) = 30.0 L $H_2$.

## B. Limiting reactant and percent yield

**3.** 10.0 g of Al react with 20.0 g of $Cl_2$: $2Al + 3Cl_2 \rightarrow 2AlCl_3$. Which is limiting, and how many grams of $AlCl_3$ can form? How much of the excess reactant is left?

| (mol) | 2Al | $3Cl_2$ | $2AlCl_3$ |
|---|---|---|---|
| **Before** | 0.371 | 0.282 | 0 |
| **Change** | -0.188 | -0.282 | +0.188 |
| **After** | 0.183 | 0 | 0.188 |

Limiting reactant: $Cl_2$ (it gives the smallest amount of reaction, 0.0940 "equation units").
$AlCl_3$ produced: 0.188 mol = 25.1 g
Al left over: 0.183 mol = 4.93 g

**4.** 6.54 g of Zn react with 0.150 mol of HCl: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. Find the liters of $H_2$ at STP that can form and the grams of the excess reactant left.

| (mol) | Zn | 2HCl | $ZnCl_2$ | $H_2$ |
|---|---|---|---|---|
| **Before** | 0.100 | 0.150 | 0 | 0 |
| **Change** | -0.0750 | -0.150 | +0.0750 | +0.0750 |
| **After** | 0.0250 | 0 | 0.0750 | 0.0750 |

Limiting reactant: HCl (it gives the smallest amount of reaction, 0.0750 "equation units").
$H_2$ produced: 0.0750 mol = 1.68 L
Zn left over: 0.0250 mol = 1.64 g

**5.** A student burns 4.86 g of Mg in air: $2Mg + O_2 \rightarrow 2MgO$. The white powder has a mass of 7.60 g.

**a.** Calculate the theoretical yield of MgO.

4.86 g Mg × (1 mol Mg / 24.31 g) × (2 mol MgO / 2 mol Mg) × (40.31 g MgO / 1 mol) = 8.06 g MgO

**Answer (units & sig figs):** **8.06 g MgO**

**b.** Calculate the percent yield and suggest a reason it differs from 100 %.

7.60 / 8.06 × 100 = 94.3 %. Some MgO smoke escaped, or some Mg formed $Mg_3N_2$ with nitrogen in the air.

**Answer (units & sig figs):** **94.3 %**

## C. Lab data

**6.** A student mixes 2.12 g of $Na_2CO_3$ (in water) with excess $CaCl_2$ solution: $Na_2CO_3$ (aq) + $CaCl_2$ (aq) → $CaCO_3$ (s) + 2NaCl (aq). The precipitate is filtered, dried and weighed. Data: filter paper 0.86 g; filter paper + dry solid 2.76 g.

(a) Mass of $CaCO_3$ collected: 2.76 − 0.86 = 1.90 g.
(b) Theoretical: 2.12 g $Na_2CO_3$ × (1 mol $Na_2CO_3$ / 105.99 g) × (1 mol $CaCO_3$ / 1 mol $Na_2CO_3$) × (100.09 g $CaCO_3$ / 1 mol) = 2.00 g $CaCO_3$
(c) Percent yield: 1.90 / 2.00 × 100 = 95.0 %

## D. Think

**7.** A recipe makes 24 cookies from 2 cups of flour. You have 5 cups of flour but only 3 eggs, and the recipe needs 1 egg per 8 cookies. How many cookies can you make, and what is the limiting ingredient? Relate this to a chemical reaction.

Eggs: 3 × 8 = 24 cookies. Flour: 5 / 2 × 24 = 60 cookies. Eggs limit, so 24 cookies. In a reaction, the reactant that gives the least product is limiting, just as the eggs are.

**8.** Why must you convert to moles before deciding which reactant is limiting?

The equation compares particles (moles), not grams; equal masses of two substances are different numbers of particles.

**9.** $N_2 + 3H_2 \rightarrow 2NH_3$. If 4 mol $N_2$ and 9 mol $H_2$ are mixed, the limiting reactant is…

A) $N_2$ **B) $H_2$** C) $NH_3$ D) neither

**10.** Which quantity cannot be negative or over 100 % in a lab report?

**A) percent yield** B) actual yield minus theoretical C) mole ratio D) Change in a BCA table

## Unit 6 review

**11.** Write the complete plan (in words) for solving "grams of A → grams of B". Then say what changes for "liters of gas A → grams of B".

Balance; grams A ÷ molar mass A = mol A; × (coefficient B / coefficient A) = mol B; × molar mass B = grams B. For liters of A, replace the first step with liters ÷ 22.4 L/mol (at STP).

**12.** Describe how you find the limiting reactant and how you use it to find the product and the leftover.

Convert each reactant to moles and divide by its coefficient; the smallest quotient is limiting. Use the limiting reactant's moles with the mole ratio to get product; subtract what was used from the excess reactant's starting moles to find the leftover.

### Sources & further reading

- PhET – Reactants, Products and Leftovers simulation (University of Colorado Boulder) — https://phet.colorado.edu/en/simulations/reactants-products-and-leftovers
- LibreTexts – Introductory Chemistry (CK-12), Chapter 12: Stoichiometry — https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Introductory_Chemistry_%28CK-12%29/12%3A_Stoichiometry
- LibreTexts – Chemistry 2e (OpenStax), 4.4 Reaction Yields — https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_2e_(OpenStax)/04%3A_Stoichiometry_of_Chemical_Reactions/4.04%3A_Reaction_Yields